**Sheet 12**

# Global tests and retained records

1. For $\Psi_\phi = (|000\rangle + e^{i\phi}|111\rangle)/\sqrt{2}$ compute the one- and two-qubit marginals. Find the means of $B_x = |000\rangle\langle 111| + |111\rangle\langle 000|$ and $B_y = -i|000\rangle\langle 111| + i|111\rangle\langle 000|$. Give a phase-matched witness with mean one for every $\phi$, against zero for the incoherent mixture.
2. In the two-dimensional span of calibrated orthogonal outputs $\chi_0, \chi_1$, form the equal superposition $\chi$. Compute the eigenvalues of $|\chi\rangle\langle\chi| - |\chi_0\rangle\langle\chi_0|$ and its trace norm. Compare with the robust theorem's separation at $\eta = 0.1$ and $\eta = 1/2$. Why must calibration explicitly place each output in its labelled set $S_i$? This is a finite application of the cited theorem, not its general proof.
3. Copy the three-qubit chain's last record to a fourth qubit, producing $(|0000\rangle + |1111\rangle)/\sqrt{2}$. Undo only the two original CNOTs, leaving the fourth record untouched. Compute the final system $+x$ probability. Starting afresh from the four-qubit state, in which order must all three couplings be reversed to restore the original interference? Why is undoing the third-to-fourth coupling ineffective after its control record has already been reset?